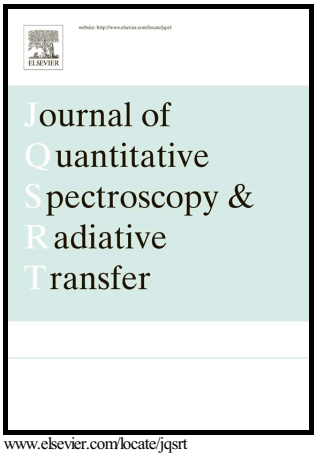
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Radiative association of He(2^3P) with lithium cations: $\Pi \rightarrow \Sigma$ processes

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Abstract

The radiative association processes originating in the $1^3\Pi$ continuum of the He$(2^3\mathrm{P})$ + Li^+ collisional system are investigated in this study. The calculations of the dynamic collision processes are based on highly accurate state-of-the-art *ab initio* calculations of the potential energy functions for the $1^3\Pi$ and the three lowest $^3\Sigma$ states of $HeLi^+$ and the associated transition dipole-moment functions. Cross-sections for the spontaneous and stimulated radiative association processes are calculated as functions of collision energy. The corresponding rate coefficients characterizing the efficiency of the formation of the molecular ion in its $a^3\Sigma^+$, $b^3\Sigma^+$, and $c^3\Sigma^+$ states from the initial $1^3\Pi$ state are obtained over a wide range of temperatures. At very low temperatures the $1 \rightarrow b$ process has a maximum rate constant value of about 7.9×10^{-13} cm^{3}s^{-1}, whereas process $1 \rightarrow a$ reaches its maximum value of 2.0×10^{-13} cm^{3}s^{-1} at a temperature of about 500 K. Altogether the three radiative association processes investigated here can be considered as the continuum-to-bound state radiative transition part of the total quenching of the initial collision channel.

Key words: metastable helium, molecular ions, rates of formation, radiative association

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1. Introduction

Due to the relatively large helium abundance in the Universe, helium lines are observed in a wide variety of stellar objects, such as for example in early-type stars [1], in gaseous nebulae [2] or Quasy-Stellar-Objects (QSO's) [3]. A number of experimental and theoretical studies were thus performed in the past to obtain accurate information on the different helium states for interpreting these observations [4, 5, 6, 7, 8]. On the theoretical side these include studies of electron excitations from the $1^1\mathrm{S}$ ground state to the lowest metastable triplet states of helium ($2^3\mathrm{S}$ and $2^3\mathrm{P}$) [9], or from $1^1\mathrm{S}$ to the low-lying metastable singlet states ($2^1\mathrm{S}$ and $2^1\mathrm{P}$) [10], as well as excitations between these metastable states [11]. However, apart from electron excitations, depopulation processes of these helium metastables by collisions with atomic ions, especially with H^+ or He^+ and even with Li^+, are also of some interest in the modelling attempts of stellar atmospheres [12, 13] or of planetary nebulae [14, 15, 16]. In this context depopulations of metastable $\mathrm{He}(2^3\mathrm{S})$ and $\mathrm{He}(2^1\mathrm{S})$ in collisions with hydrogen, helium, and lithium ions by radiative associations [17, 18, 19, 20, 21] and also by non-radiative inelastic processes [22, 23] were extensively studied. Depopulations of $\mathrm{He}(2^3\mathrm{P})$ by collisions with hydrogen and helium ions leading via radiative association to the formation of HeH^+ [24] or He_2^+ [18], respectively, were investigated as well as most recently by collisions with lithium ions [25]. Whereas depopulation processes of the He metastables in collisions with hydrogen and helium ions can contribute to the helium chemistry in the relevant interstellar environments, the corrresponding processes with lithium ions are probably irrelevant in this respect due to the very small primordial Li/He element abundance ratio. These processes can however affect possibly the low Li^+ abundance at high redshifts in the early Universe.

In this respect, as an extension of the recent study of Ref. [25] in which the radiative association processes originating in the triplet-Sigma continuum of the collisional system $\mathrm{He}(2^3\mathrm{P})$ + Li^+ were considered, the present study puts its focus on the corresponding radiative association processes originating in the continuum of the lowest $^3\Pi$ state. The considered processes induced by dipole-moment continuum-bound transitions are described as:

$$1^3\left[\mathrm{He}(2^3\mathrm{P}) + \mathrm{Li}^+\right] \rightarrow \mathrm{HeLi}^+(a^3\Sigma^+) + h\nu \quad (1 \rightarrow a), \tag{1}$$

$$1^3\left[\mathrm{He}(2^3\mathrm{P}) + \mathrm{Li}^+\right] \rightarrow \mathrm{HeLi}^+(b^3\Sigma^+) + h\nu \quad (1 \rightarrow b), \tag{2}$$

$$1^3\left[\mathrm{He}(2^3\mathrm{P}) + \mathrm{Li}^+\right] \rightarrow \mathrm{HeLi}^+(c^3\Sigma^+) + h\nu \quad (1 \rightarrow c), \tag{3}$$

where $1^3[\ldots]$ means the corresponding collisional continuum of the $1^3\Pi$ electronic states of $HeLi^+$.

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The three reactions in this scheme describe the so-called two-state processes (two electronic states are involved). According to general experience they are expected to have larger efficiencies compared to one-state processes due to a larger energy gain by the spontaneous photon emission in the transitions from the initial to the target state and also a potentially better overlap between the continuum and bound wave functions. Reaction (3), on the other hand, is an example for a two-state process where the general expectation does not hold. The initial $1^3\Pi$ state and the $c^3\Sigma^+$ target state are degenerate at the dissociation limit with identical dissociation products $Li^+(1^1S)$ + $He(2^3P)$. Moreover, since the potential curves of the two states are close to each other over the entire distance range, the energy gain in this process is rather small. Inspection of the relevant potential energy curves obtained previously [26] indicates that the largest energy gain occurs for reaction (1) in the above scheme. In this reaction the initial reaction channel $Li^+(1^1S)$ + $He(2^3P)$ is connected to the $a^3\Sigma^+$ state of $HeLi^+$ dissociating to $Li(2^2S)$ + $He^+(1^2S)$ whereas in reaction (2) the initial channel is connected to the $b^3\Sigma^+$ state which dissociates to $He(2^3S)$ + $Li^+(1^1S)$. In this respect, reactions (1) and (2) can be viewed to be related to charge-transfer or He de-excitation processes, respectively.

2. Methods

The quantum dynamics calculations were performed making use of potential energy curves of the $a^3\Sigma^+$, $b^3\Sigma^+$, $c^3\Sigma^+$, and $1^3\Pi$ electronic states of $HeLi^+$ and the transition dipole moments between the 1 and a, b or c states, where the corresponding *ab initio* interaction energies and moment-function values were taken from Soldán and Kraemer [26]. We used the one-dimensional reproducing kernel Hilbert space (RKHS) interpolation method [27, 28, 29] for numerical representation of the potential energy curves and transition moment functions. The same reciprocal-power reproducing kernel Hilbert space (RP-RKHS) representation as in Ref. [25] was used for the potential energy curves of the a, b, and c states. For the 1-state potential energy curve we also used the RP-RKHS interpolation. In the $1^3\Pi$ state the charge-quadrupole interaction between the Li^+ and $He(2^3P)$ is repulsive and the leading long-range term is C_3/R^3, thus the interpolation was done with respect to R using RP-RKHS parameters $m = 2$ and $n = 2$. The resulting value of $C_3 = 5.07\ a_0^3\ E_h$ was extracted from the RP-RKHS extrapolation.

Calculations of cross-sections at various collision energies and rate coefficients over various temperatures form the base of our radiative-association treatment. In our computer-program suite [30] we implemented the approach to the

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formation of diatomic bound species via radiative association as reviewed by Dalgarno and coworkers [31, 32].

The total radiative-association cross-section $\sigma(E;T_\mathrm{b})$ at collision energy E in the field of black-body radiation (characterized by temperature T_b) is a sum of individual cross-sections, which correspond to allowed transitions from an initial continuum state (characterized by E and angular momenta q') to a target bound ro-vibrational state (labeled by quantum numbers v'' and q'')

$$\sigma(E;T_\mathrm{b}) = \sum_{q',v'',q''} \sigma_{q';v'',q''}(E;T_\mathrm{b})\,. \tag{4}$$

In the case of the $^3\Pi \rightarrow {}^3\Sigma^+$ transitions q represents electronic angular momentum Ω and overall angular momentum J. The electronic angular momentum $\Omega = \mid \Lambda + \Sigma \mid$,[1] where $\Lambda = 0$ and ± 1 for Σ and Π states, respectively, and $\Sigma = 0$ or ± 1 for a triplet state. The overall angular momentum can then have values $J = \Omega, \Omega+1, \Omega+2, \ldots$ The dipole-moment $^3\Pi \rightarrow {}^3\Sigma^+$ transitions are partly governed by selection rule $\triangle\Omega = \Omega' - \Omega'' = \pm 1$ ($\triangle\Lambda = \Lambda' - \Lambda'' = \pm 1$ and $\triangle\Sigma = \Sigma' - \Sigma'' = 0$), which leads to three $\Omega-$transitions ($\Omega-$subprocesses) $\Omega' = 0 \rightarrow \Omega'' = 1$, $1 \rightarrow 0$, and $2 \rightarrow 1$. Then the sum in Eq. (4) can also be expressed as a sum over different $\Omega-$transitions

$$\sigma(E;T_\mathrm{b}) = \sigma^{0\rightarrow 1}(E;T_\mathrm{b}) + \sigma^{1\rightarrow 0}(E;T_\mathrm{b}) + \sigma^{2\rightarrow 1}(E;T_\mathrm{b}), \tag{5}$$

where we grouped together the individual cross-sections belonging to a certain $\Omega-$subprocess

$$\sigma^{\Omega'\rightarrow\Omega''}(E;T_\mathrm{b}) = \sum_{J',v'',J''} \sigma_{J',\Omega';v'',J'',\Omega''}(E;T_\mathrm{b})\,. \tag{6}$$

The individual cross-sections can be expressed as

$$\sigma_{q';v'',q''}(E;T_\mathrm{b}) = \frac{1}{4\pi\epsilon_0}\frac{64}{3}\frac{\pi^5}{c^3\,k^2}\,p\,\nu^3_{E;v'',q''}\,S_{q',q''}\,|M_{E,q';v'',q''}|^2 \quad \times \\ \times \left[\frac{1}{1-\exp(-h\nu_{E;v'',q''}/k_\mathrm{B}T_\mathrm{b})}\right], \tag{7}$$

where c is the speed of light in vacuum, $k^2 = 2\mu E/\hbar^2$, μ is the reduced mass of the molecular ion ($\mu = 4645.856524$ a.u. for ^{4}He^{7}Li$^+$), p is the $\Omega-$distinguished

[1] In this work, it is convenient to use this definition of Ω. Alternative definition would be $\Omega = \Lambda + \Sigma$.

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probability of approach in the initial electronic state ($p = 2/9$ for all three processes) and $\nu_{E;v'',q''}$ is the emitted photon frequency, $h\nu_{E;v'',q''} = E + \triangle E - E_{v'',q''}$, where $\triangle E_{1\to c} = 0$ eV, $\triangle E_{1\to b} = 1.1444818$ eV, and $\triangle E_{1\to a} = 1.7684229$ eV [33]. Hönl-London coefficients $S_{q',q''}$ collected in Table 1 then reflect the complete set of selection rules for the electric dipole $^3\Pi \to {}^3\Sigma$ transitions [34].

Table 1: Hönl-London coefficients $S_{q',q''}$ for electric dipole transitions for the $^3\Pi \to {}^3\Sigma$ process

process $\Omega' \to \Omega''$	$\triangle J = -1$ P-branch	$\triangle J = 0$ Q-branch	$\triangle J = +1$ R-branch
$0 \to 1$	$J' + 2$	$2J' + 1$	$J' - 1$
$1 \to 0$	J'	$2J' + 1$	$J' + 1$
$2 \to 1$	$\frac{(J'-1)J'}{J'+1}$	$\frac{(2J'+1)(J'+2)(J'-1)}{J'(J'+1)}$	$\frac{(J'+1)(J'+2)}{J'}$

$$M_{E,q';v'',q''} = \chi_{E,q'} T^1_k(\mathbf{d}) \psi_{v'',q''} = \int_0^\infty \chi_{E,q'}(R)\, T^1_k(\mathbf{d})\, \psi_{v'',q''}(R)\, \mathrm{d}R \qquad (8)$$

is the matrix element of the transition dipole-moment vector $\mathbf{d}(R)$ in the spherical tensor representation between the initial continuum radial wave function $\chi_{E,q'}(R)$ and the final bound-state radial wave function $\psi_{v'',q''}(R)$ ($k = 1$ for the $\Pi \to \Sigma$ transitions, $T^1_1(\mathbf{d}) = \frac{1}{\sqrt{2}}\left[-d_x(R) + i\, d_y(R)\right]$ and $d_x(R) = d_y(R)$ for diatomic molecules, thus in our case $|M_{E,q';v'',q''}|^2 = |\chi_{E,q'} d_x \psi_{v'',q''}|^2$). The above formula (7) is derived for the normalized bound-state wave function

$$\int_0^\infty |\psi_{v'',q''}(R)|^2\, \mathrm{d}R = 1 \qquad (9)$$

and the energy-normalized continuum wave function

$$\chi_{E,q'}(R) \sim \left(\frac{2\mu}{\hbar^2 \pi k}\right)^{1/2} \sin\left[kR - \frac{1}{2}\, J'\pi + \delta_{q'}(E)\right] \qquad (10)$$

for $R \to \infty$, where $\delta_{N'}(E)$ is the phase shift. To determine the radial wave functions $\chi_{E,q'}(R)$ and $\psi_{v'',q''}(R)$ we use numerical integration [35, 36] of the

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corresponding Schrödinger equations

$$\left[-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2}{\mathrm{d}R^2}+\frac{\hbar^2}{2\mu}\frac{J'(J'+1)-\Omega'^2}{R^2}+U_C(R)\right]\chi_{E,q'}(R)= \\ = E\,\chi_{E,q'}(R), \tag{11}$$

$$\left[-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2}{\mathrm{d}R^2}+\frac{\hbar^2}{2\mu}\frac{J''(J''+1)-\Omega''^2}{R^2}+U_B(R)\right]\psi_{v'',q''}(R)= \\ = E_{v'',q''}\,\psi_{v'',q''}(R), \tag{12}$$

which include potential energy curves $U_C(R)$ and $U_B(R)$ supporting the initial continuum state E, q' and the final bound state v'', q'', respectively.

We consider the $\Omega-$independent potential energy curves and transitions dipole-moment functions from Soldán and Kraemer [26] to be good approximations. Making use of the Airy-function boundary condition at the outermost classical turning point [37] and the uniform semiclassical method [38, 39], the positions and tunnelling widths of resonances were calculated employing the computer program LEVEL 7.7 [40].

The Hund's case (a) Spin-Less Approximation (SLA) [41] can then be introduced by setting $\Sigma' = \Sigma'' = 0$, which results in $\Omega' = \mid \Lambda' \mid = 1$ for the Π state and $\Omega'' = \mid \Lambda'' \mid = 0$ for the Σ state. The total radiative-association cross-section $\sigma(E;T\mathrm{b})$ is then approximated

$$\sigma(E;T_\mathrm{b}) \approx \sigma_\mathrm{SLA}(E;T_\mathrm{b}) = 3\,\sigma^{1\rightarrow 0}(E;T_\mathrm{b}). \tag{13}$$

The total rate coefficients for formation of a molecule by (spontaneous, $T_\mathrm{b} = 0$ K, or stimulated, $T_\mathrm{b} > 0$ K) radiative association at a given temperature T are obtained by averaging the corresponding cross-sections $\sigma(E;T_\mathrm{b})$ over a Maxwellian velocity distribution of the collisional continuum

$$\alpha(T;T_\mathrm{b}) = \left(\frac{8}{\mu\pi}\right)^{1/2}\left(\frac{1}{k_\mathrm{B}\,T}\right)^{3/2}\int_0^\infty E\,\sigma(E;T_\mathrm{b})\,e^{-E/k_\mathrm{B}T}\mathrm{d}E, \tag{14}$$

where T_b is the background temperature. Similarly, the partial rate coefficients $\alpha^{\Omega'\rightarrow\Omega''}(T;T_\mathrm{b})$ and approximated rate coefficient $\alpha_\mathrm{SLA}(T;T_\mathrm{b})$ can be introduced. For the numerical evaluation of integral (14) the approach outlined in Ref. [30] is followed. The Kooij function is used in order to represent analytically the total rate coefficients

$$f(T) = \alpha\,\left(\frac{T}{300}\right)^{\beta}\,e^{-\gamma/T} \tag{15}$$

where α, β, and γ are fitting parameters.

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3. Results and discussion

The potential energy curve of the $1^3\Pi$ state of $HeLi^+$ is shown in Fig. 1 together with those of the three lowest $^3\Sigma^+$ states acting as target states in this study, whereas Fig. 2 presents the corresponding transition dipole-moment functions for the three association processes $1 \rightarrow a$, $1 \rightarrow b$, and $1 \rightarrow c$. According to our earlier calculations [26] the potential energy curve of the initial $1^3\Pi$ state exhibits a small barrier with maximum of 64.9 cm^{-1} at 18.99 a_0 due to the electrostatic charge-quadrupole interaction. This barrier can support two rotation-less resonances.

In Table 2 some basic data are collected specifying the numbers of bound and quasi-bound states, which are supported by the electronic states considered here. The shapes of the potential energy curves of the target states are quite different. Whereas the $a^3\Sigma^+$ state has a fairly deep potential well with $\mathcal{D}_e$=8328.0 cm^{-1} which can hold a large number of well localized ro-vibrational states, the shapes of the potentials of the other two $b^3\Sigma^+$ and $c^3\Sigma^+$ states are shallower (with $\mathcal{D}_e$=872.4 cm^{-1} and $\mathcal{D}_e$=1438.7 cm^{-1} for the two states) reflecting the weak and slowly varying long-range induction and charge-quadrupole interaction forces between the $Li^+(1^1S)$ ion and helium metastables $He(2^3S)$ and $He(2^3P)$, respectively. However, since these weak interaction potentials are rather wide they can still accommodate a fairly large number of delocalized ro-vibrational states as indicated in Table 2. The different characters especially between the a- and b-states are also demonstrated by the different behaviour of the transition dipole-moment functions responsible for the $1 \rightarrow a$ and $1 \rightarrow b$ association processes shown in Fig. 2. The $1 \rightarrow a$ function converges to zero with increasing internuclear distance whereas the $1 \rightarrow b$ function converges to a constant non-zero value characteristic for the de-excitation process $He(2^3P) \rightarrow He(2^3S)$.

Cross-sections for the three spontaneous radiative association processes considered here are illustrated for the $\Omega-$subprocess $1 \rightarrow 0$ as functions of collision energy in Fig. 3. The curve for the process $1 \rightarrow b$ has the usual shape known for similar processes. It starts with its maximum values at lowest energies, decreases steadily with increasing collision energy and has finally a rapid drop-off at energies higher than 10^{-1} eV. Over the entire energy range the curve shows few wider resonances with small resonance peaks gathering in the energy region from 10^{-2} eV to 10^{-1} eV. On the other hand, the shapes of the cross-section curves for processes $1 \rightarrow a$ and $1 \rightarrow c$ at energies below 10^{-2} eV are totally different. In this low energy region both curves are rapidly increasing until they reach their maximum values at energies above 10^{-2} eV. Before reaching their maxima both curves show an identical rich resonance structure which means that the initial complex

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formation taking place in the continuum of the $1^3\Pi$ state potential is identical for both processes at this low energy. After reaching their maximum values the further energy dependence of the $1 \rightarrow a$ and $1 \rightarrow c$ processes follows the usual pattern, i.e. decrease with increasing energy and rapid drop-off at higher energies. Actually the drop-off of the $1 \rightarrow c$ process occurs almost at the same energy as that for $1 \rightarrow b$, whereas the $1 \rightarrow a$ curve falls off at an energy of about 1 eV. The cross-sections for the $1 \rightarrow c$ process are as expected several orders of magnitude smaller compared to the other two processes. In this case, possible energy gains by the spontaneous photon emissions are very small.

The reason for the different character of the three cross-section curves at low collision energies should be seen in the interplay between the influence of the $1^3\Pi$ barrier, which suppresses the values of the initial continuum wave function in front of the barrier, and the different asymptotic behaviour of the relevant transition dipole-moment curves (Fig. 2), which change the corresponding continuum-bound Franck-Condon overlaps in a nontrivial manner. At low collision energies the integration in Eq. (8) has to be extended to very large distances, particularly for vibrationally highly excited target bound states. At these large distances (beyond the $1^3\Pi$ barrier maximum) the transition dipole moments for the $1 \rightarrow a$ and $1 \rightarrow c$ processes go asymptotically to zero, whereas for $1 \rightarrow b$ it converges to a non-zero constant value. The combined effect of the barrier and transition dipole moments is nicely illustrated in Fig. 4, where the transition dipole-moment matrix elements $|\chi_{E,q'} d_x \psi_{v'',q''}|$ are shown as functions of vibrational quantum number v'' ($\Omega' = 1, J' = 1$ and $\Omega'' = 0, J'' = 0$) for the $1 \rightarrow a$ and $1 \rightarrow b$ processes at two different collision energies $E = 0.001$ eV and $E = 0.075$ eV. It shows that at $E = 0.001$ eV this matrix element for the $1 \rightarrow a$ process is much smaller than it is for the $1 \rightarrow b$ process. On the other hand, at $E = 0.075$ eV these matrix elements are comparable for both the processes, and after summing over all quantum numbers (see Eq. (4)) the $1 \rightarrow a$ process becomes slightly more favourable. This is further emphasised in Fig. 5, where the corresponding contributions to the transition dipole-moment matrix elements $\chi_{E,q'} d_x \psi_{v'',q''}$ are shown when integrated from 0 to 19 a_0 and from 19 a_0 to infinity (sum of these contributions gives the entire matrix element; 19 a_0 is very near the $1^3\Pi$ barrier maximum). The consequence is that for the $1 \rightarrow a$ and $1 \rightarrow c$ processes the cross-section curves start from very small values at low collision energies whereas the $1 \rightarrow b$ cross-section curve starts at quite high values.

Another peculiar feature in the cross-sections (see Fig. 3) is the lack of profound resonance structure in the process $1 \rightarrow b$ at low energies whereas it is so clearly visible in the other two processes. In fact, this is just a consequence: a)

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of our visualization method (only resonances with widths larger than 0.01 cm^{-1} are shown, because the figure would otherwise become very messy), and b) of the shape of the transition dipole-moment function for the $1 \rightarrow b$ process. This transition dipole-moment function can be represented as $d_x(R) = f(R) + f_\infty$, where $f_\infty = -2.537812\ ea_0$ is a constant and the function $f(R)$ converges to zero for large R (beyond the $1^3\Pi$ potential barrier maximum, see dashed blue curve in Fig. 2). This divides each $|\chi_{E,q'} d_x \psi_{v'',q''}|^2$ into the sum of three terms: $|\chi_{E,q'} f \psi_{v'',q''}|^2 + f_\infty^2\ |\chi_{E,q'} \psi_{v'',q''}|^2 + 2 f_\infty \chi_{E,q'} f \psi_{v'',q''}$. Fig. 6 shows the contributions of these three terms to the cross-sections of the $\Omega-$subprocess $1 \rightarrow 0$. The resonance structure is clearly seen in the two terms involving the function $f(R)$, however it is almost missing in the f_∞^2 term contribution. This is caused by the shape of the function $f(R)$, which is quite similar to the transition dipole-moment functions of the $1 \rightarrow a$ and $1 \rightarrow c$ processes, and similarly it emphasizes the region in front of the $1^3\Pi$ barrier that is so crucial for shape resonances. In the end, it effectively "widens" the neighbourhood of many resonances so that they appear with larger width than 0.01 cm^{-1} and become then visible in our figure.

Rate coefficients for the spontaneous radiative associations are determined from the cross-sections and are shown in Fig. 7 as functions of temperature (for comparison the rate coefficients for the corresponding $\Sigma \rightarrow \Sigma$ processes, calculated in Ref. [25], are also shown in Fig. 7). Numerical values evaluated at selected temperatures are provided in Table 3 and Table 4. Table 3 lists the partial, total and approximated rate coefficients for process $1 \rightarrow a$ at temperatures up to 1000 K whereas Table 4 presents a list of total and approximated rate coefficients for all the three processes considered here at temperatures up to 500,000 K. Comparison of the rate coefficient values for the $1 \rightarrow a$ process in the two tables shows that small differences in the $\Omega-$transitions play only a role at lower temperatures (below 100 K), and that at higher temperatures the spin-less approximation ($\Omega =\mid \Lambda \mid$) appears to be valid. The shapes of the rate curves correspond to those of the underlying cross-section graphs. For processes $1 \rightarrow a$ and $1 \rightarrow c$ the curves reach their maximum values at higher temperatures whereas for $1 \rightarrow b$ the curve is steadily decreasing from its initial maximum value of almost 10^{-12} cm^{3}s^{-1} at T = 10 K. The maximum of the $1 \rightarrow a$ curve occurs at a temperature of about 500 K with a rate coefficient value of the order of 10^{-13} cm^{3}s^{-1}. Rate coefficients for the $1 \rightarrow c$ process are as expected several orders of magnitude smaller. Table 5 contains values of the Kooij-function parameters fitted to the total rate coefficients. The corresponding temperature range was divided into several intervals in order to keep the maximum relative error below 10% in the fitting procedures.

It was previously shown that stimulated radiative association by background

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radiation is only relevant for one-state processes, or more generally, for processes with $\triangle E = 0$ [17]. The rate coefficients for the stimulated radiative association process $1 \rightarrow c$ are shown for different black-body radiation temperatures T_b in Fig. 8. Whereas in the previous study, in which the radiative association processes originating in the triplet-Sigma continuum were considered, the $c \rightarrow c$ stimulated rate coefficients ran practically parallel with constant distances to the steadily decreasing curve of the spontaneous process [25], the situation for the $1 \rightarrow c$ stimulated rate coefficients of the present study is somewhat different due to the different character of the $1 \rightarrow c$ process. As the curve for the spontaneous process, the stimulated curves ($T_\mathrm{b} > 0$) start at low rate coefficient values and run through maximum values with varying distances between the curves.

Table 2: Process characteristics for radiative association of He(2^3P) with lithium ions ($\mathcal{D}_0$ and $\triangle E$ are in cm^{-1}).[a]

process	$\Omega' \rightarrow \Omega''$	n_tot	n_r	n_w	n_n	$\mathcal{D}_0$	n_v	n_rv	$\triangle E$
$1 \rightarrow a$	$0 \rightarrow 1$	368	368	191	177	8182.9	61	4179	14263.2
	$1 \rightarrow 0$	367	367	190	177	8182.2	61	4176	14263.2
	$2 \rightarrow 1$	364	364	190	174	8182.9	61	4179	14263.2
$1 \rightarrow b$	$0 \rightarrow 1$	368	338	164	174	829.8	33	1289	9230.9
	$1 \rightarrow 0$	367	337	163	174	829.6	33	1289	9230.9
	$2 \rightarrow 1$	364	334	163	171	829.8	33	1289	9230.9
$1 \rightarrow c$	$0 \rightarrow 1$	368	368	191	177	1397.5	55	2722	0
	$1 \rightarrow 0$	367	367	190	177	1397.4	55	2722	0
	$2 \rightarrow 1$	364	364	190	174	1397.5	55	2722	0

[a]The MRCI potential energy curves of Soldán and Kraemer [26] are used for all the electronic states. According to the definitions of Augustovičová et al. [30]: n_tot is the total number of orbital resonances supported by the initial electronic state, from which n_r is the number of symmetry-allowed resonances, i.e. the number of resonances which have a bound partner in the corresponding target electronic state to which transitions are allowed by the dipole-moment selection rules, n_w is the number of wide resonances, and n_n is the number of narrow resonances; $\mathcal{D}_0$ is the dissociation energy of the target electronic state from the ground ($v'' = 0, J'' = \Omega''$) ro-vibrational state, n_v is the number of bound vibrational ($J'' = \Omega''$) states of the target electronic state, n_rv is the total number of bound ro-vibrational states of the target electronic state, and $\triangle E$ is the energy difference between the dissociation levels of the potentials.

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Table 3: Rate coefficients ($cm^3\,s^{-1}$) for the spontaneous radiative association $1 \rightarrow a$ process of the $HeLi^+$ molecular ion.[a]

T (K)	$\alpha^{0\rightarrow1}$	$\alpha^{1\rightarrow0}$	$\alpha^{2\rightarrow1}$	α	α_{SLA}
5	2.24(-17)	2.51(-17)	3.15(-17)	7.90(-17)	7.52(-17)
10	8.42(-16)	8.91(-16)	9.98(-16)	2.73(-15)	2.67(-15)
20	6.04(-15)	6.20(-15)	6.55(-15)	1.88(-14)	1.86(-14)
30	1.25(-14)	1.27(-14)	1.32(-14)	3.84(-14)	3.81(-14)
50	2.37(-14)	2.39(-14)	2.46(-14)	7.22(-14)	7.18(-14)
70	3.22(-14)	3.25(-14)	3.31(-14)	9.78(-14)	9.74(-14)
100	4.17(-14)	4.19(-14)	4.25(-14)	1.26(-13)	1.26(-13)
200	5.84(-14)	5.84(-14)	5.89(-14)	1.76(-13)	1.75(-13)
300	6.49(-14)	6.48(-14)	6.51(-14)	1.95(-13)	1.94(-13)
400	6.71(-14)	6.69(-14)	6.72(-14)	2.01(-13)	2.01(-13)
500	6.72(-14)	6.71(-14)	6.73(-14)	2.02(-13)	2.01(-13)
600	6.64(-14)	6.63(-14)	6.64(-14)	1.99(-13)	1.99(-13)
700	6.51(-14)	6.50(-14)	6.51(-14)	1.95(-13)	1.95(-13)
800	6.35(-14)	6.34(-14)	6.35(-14)	1.90(-13)	1.90(-13)
900	6.19(-14)	6.18(-14)	6.19(-14)	1.86(-13)	1.85(-13)
1000	6.02(-14)	6.01(-14)	6.02(-14)	1.80(-13)	1.80(-13)

[a] $x(-y) \equiv x \times 10^{-y}$.

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Table 4: The total rate coefficients ($cm^3\ s^{-1}$) for three radiative association processes of the $HeLi^+$ molecular ion.[a]

T (K)	$1 \rightarrow a$		$1 \rightarrow b$		$1 \rightarrow c$	
	α	$\alpha_{\rm SLA}$	α	$\alpha_{\rm SLA}$	α	$\alpha_{\rm SLA}$
10	2.73(-15)	2.67(-15)	7.87(-13)	7.92(-13)	6.43(-20)	6.30(-20)
20	1.88(-14)	1.86(-14)	5.57(-13)	5.59(-13)	4.45(-19)	4.40(-19)
30	3.84(-14)	3.81(-14)	4.60(-13)	4.61(-13)	9.09(-19)	9.02(-19)
50	7.22(-14)	7.18(-14)	3.65(-13)	3.66(-13)	1.69(-18)	1.68(-18)
100	1.26(-13)	1.26(-13)	2.65(-13)	2.66(-13)	2.86(-18)	2.85(-18)
200	1.76(-13)	1.76(-13)	1.80(-13)	1.80(-13)	3.82(-18)	3.81(-18)
500	2.02(-13)	2.01(-13)	8.62(-14)	8.62(-14)	4.06(-18)	4.05(-18)
1,000	1.81(-13)	1.80(-13)	4.06(-14)	4.07(-14)	3.00(-18)	3.00(-18)
2,000	1.37(-13)	1.37(-13)	1.69(-14)	1.69(-14)	1.65(-18)	1.65(-18)
2,500	1.21(-13)	1.21(-13)	1.25(-14)	1.25(-14)	1.30(-18)	1.30(-18)
3,000	1.08(-13)	1.08(-13)	9.74(-15)	9.75(-15)	1.05(-18)	1.05(-18)
4,000	8.72(-14)	8.71(-14)	6.52(-15)	6.52(-15)	7.40(-19)	7.40(-19)
6,000	6.08(-14)	6.08(-14)	3.65(-15)	3.65(-15)	4.38(-19)	4.38(-19)
8,000	4.53(-14)	4.53(-14)	2.41(-15)	2.41(-15)	2.97(-19)	2.97(-19)
10,000	3.53(-14)	3.53(-14)	1.74(-15)	1.74(-15)	2.18(-19)	2.18(-19)
16,000	2.00(-14)	2.00(-14)	8.71(-16)	8.71(-15)	1.12(-19)	1.12(-19)
20,000	1.50(-14)	1.50(-14)	6.26(-16)	6.26(-16)	8.10(-20)	8.10(-20)
25,000	1.12(-14)	1.12(-14)	4.50(-16)	4.50(-16)	5.86(-20)	5.86(-20)
32,000	7.98(-15)	7.97(-15)	3.12(-16)	3.12(-16)	4.08(-20)	4.08(-20)
50,000	4.27(-15)	4.27(-15)	1.60(-16)	1.60(-16)	2.11(-20)	2.11(-20)
64,000	3.00(-15)	3.00(-15)	1.11(-16)	1.11(-16)	1.47(-20)	1.47(-20)
100,000	1.57(-15)	1.57(-15)	5.68(-17)	5.68(-17)	7.55(-21)	7.55(-21)
200,000	5.67(-16)	5.67(-16)	2.01(-17)	2.01(-17)	2.68(-21)	2.68(-21)
500,000	1.45(-16)	1.45(-16)	5.10(-18)	5.10(-18)	6.81(-22)	6.81(-22)

[a] $x(-y) \equiv x \times 10^{-y}$.

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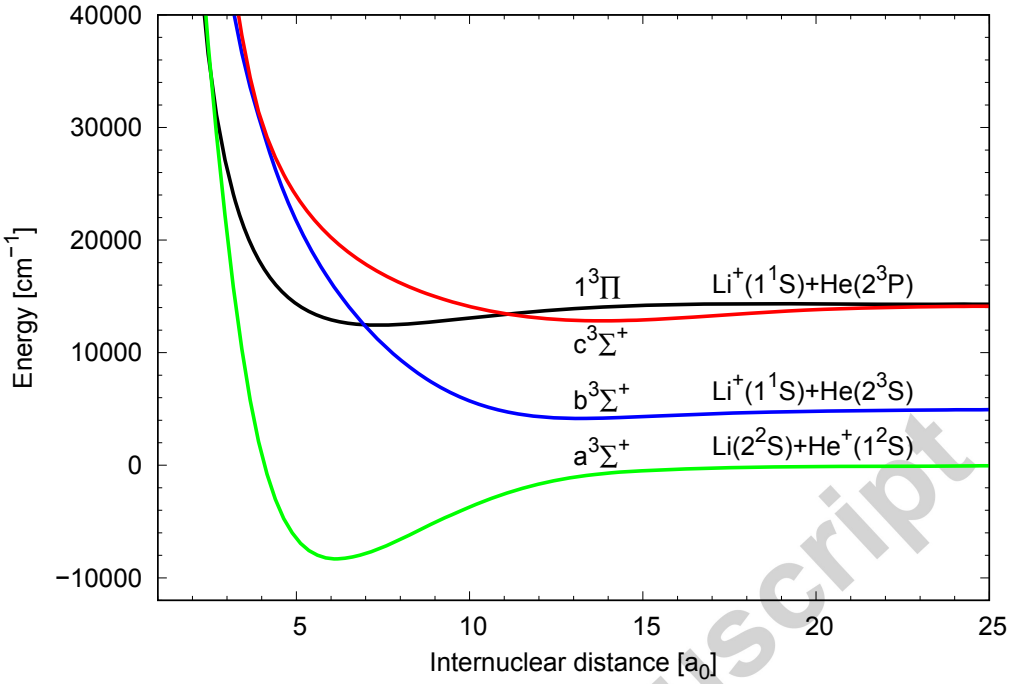


Figure 1: Potential energy curves of the $1^3\Pi$ (black), $a^3\Sigma^+$ (green), $b^3\Sigma^+$ (blue), and $c^3\Sigma^+$ (red) states of $HeLi^+$.

4. Conclusion

The radiative association processes originating in the $1^3\Pi$ continuum of the $He(2^3P)$ + Li^+ collisional system have been investigated in this study. The calculations of the dynamic collision processes are based on highly accurate state-of-the-art *ab initio* calculations of the potential energy functions for the $1^3\Pi$ and the three lowest $^3\Sigma$ states of $HeLi^+$ and the associated transition dipole-moment functions obtained previously [26]. Cross-sections for the spontaneous and stimulated radiative association processes have been calculated as functions of collision energy. The corresponding rate coefficients characterizing the efficiency of the formation of the molecular ion in its $a^3\Sigma^+$, $b^3\Sigma^+$, and $c^3\Sigma^+$ states from the initial $1^3\Pi$ state have been obtained over a wide range of temperatures.

The results of the present calculations together with those of the previous study of Ref. [25] can also be interpreted in terms of radiative depopulations of metastable $He(2^3P)$ in collisions with Li^+ ions (restricting the processes to $^3\Sigma^+$ symmetry). In this context, they can be directly compared with those which

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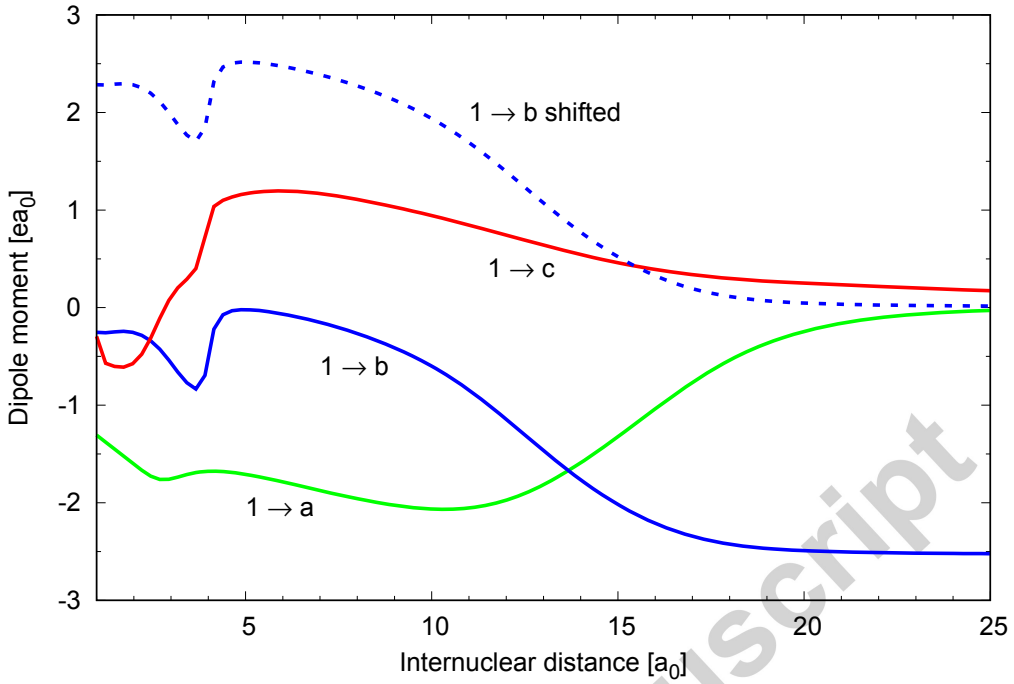


Figure 2: Components d_x of the transition dipole moment for the $1 \rightarrow a$ (green), $1 \rightarrow b$ (blue), and $1 \rightarrow c$ (red) processes in $HeLi^+$. Dashed blue line represents function $f(R)$ from the decomposition $d_x(R) = f(R) + f_\infty$ in the $1 \rightarrow b$ case, where $f_\infty = -2.537812\ ea_0$ is a constant.

were recently obtained for the radiative depopulation of $He(2^3S)$ in collisions with Li^+ ions [19]. The most efficient depopulation process in that study, the radiative association formation of the $a^3\Sigma^+$ state, turned out to have a maximum rate coefficient value of about $1.2{\times}10^{-14}$ cm^3s^{-1} at T = 10 K. In the case of radiative depopulations of metastable $He(2^3P)$ in collisions with Li^+ ions the situation is more complex. Depopulations originating in the triplet-Sigma continuum of the collisional system $He(2^3P)$ + Li^+ are less efficient compared to those originating in the continuum of the $1^3\Pi$ state. Among these, however, the $1 \rightarrow b$ process is dominating up to a temperature of about 200 K, with an initial maximum value of about $8{\times}10^{-13}$ cm^3s^{-1} whereas at higher temperatures $1 \rightarrow a$ dominates with a difference in favor of the $1 \rightarrow a$ process of about one order of magnitude.

In astrophysical context, the processes considered here can be viewed to affect possibly the low Li^+ abundance at high redshifts in the early Universe in addition to the previously studied radiative and non-radiative depopulation processes of

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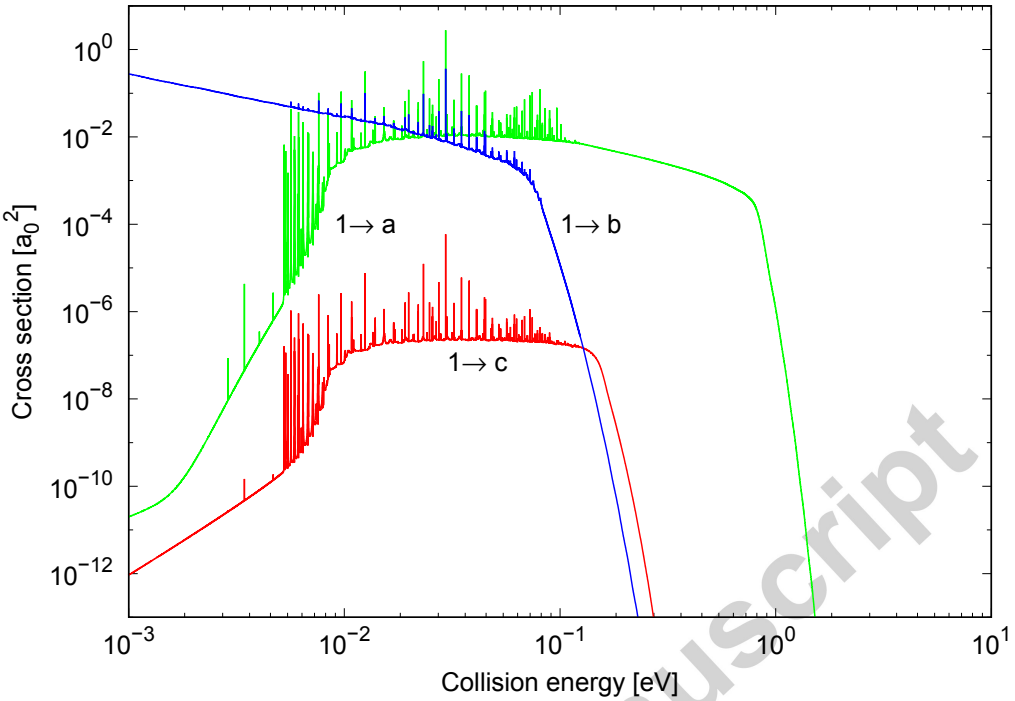


Figure 3: Cross-sections $\sigma^{1\to0}(E;0)$ for spontaneous radiative association of $HeLi^+$. Only those resonances with a width larger than 0.01 cm^{-1} are shown.

metastable He in collisions with lithium ions.

Conflict of interest

The authors declare no conflict of interest.

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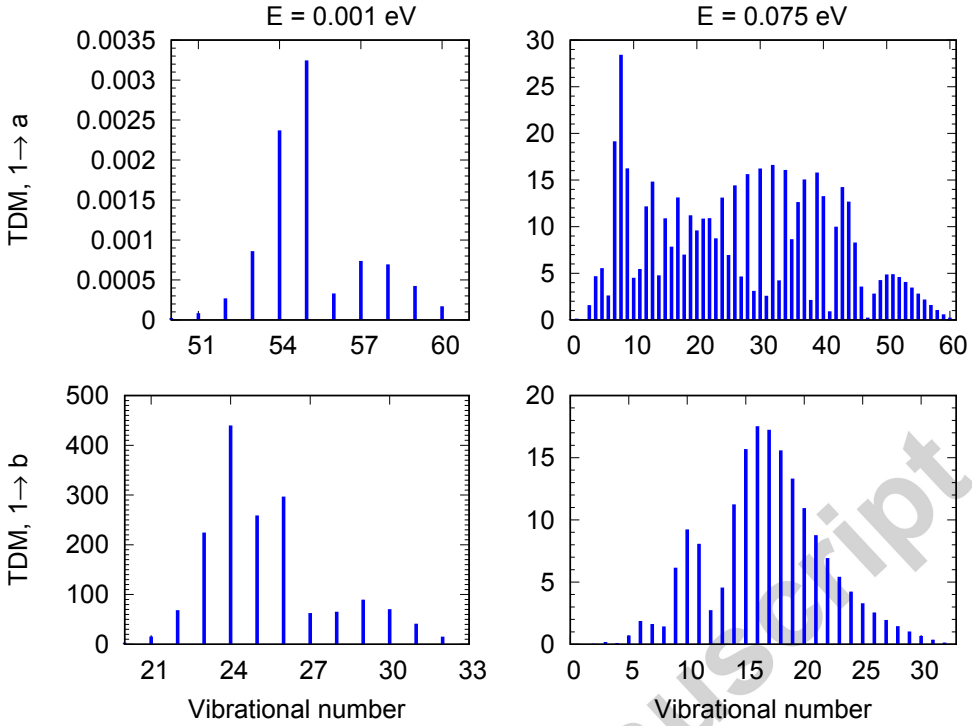


Figure 4: Comparison of the transition dipole-moment matrix elements $|\chi_{E,q'} d_x \psi_{v'',q''}|$ depending on vibrational quantum number v'' ($\Omega' = 1, J' = 1$ and $\Omega'' = 0, J'' = 0$) for the $1 \rightarrow a$ (upper panels) and $1 \rightarrow b$ (lower panels) processes at two different collision energies $E = 0.001$ eV (left panels) and $E = 0.075$ eV (right panels).

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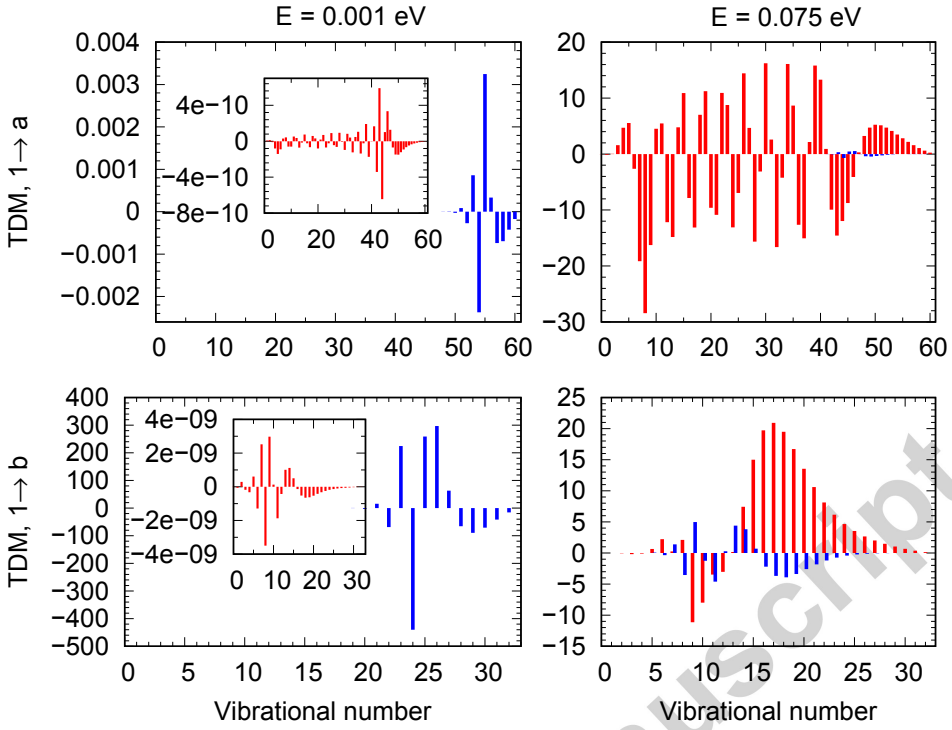


Figure 5: Contributions to the transition dipole-moment matrix elements $\chi_{E,q'} d_x \psi_{v'',q''}$: when integrated from 0 to 19 a_0 (red) and from 19 a_0 to infinity (blue) depending on vibrational quantum number v'' ($\Omega' = 1, J' = 1$ and $\Omega'' = 0, J'' = 0$) for the $1 \rightarrow a$ (upper panels) and $1 \rightarrow b$ (lower panels) processes at two different collision energies $E = 0.001$ eV (left panels) and $E = 0.075$ eV (right panels).

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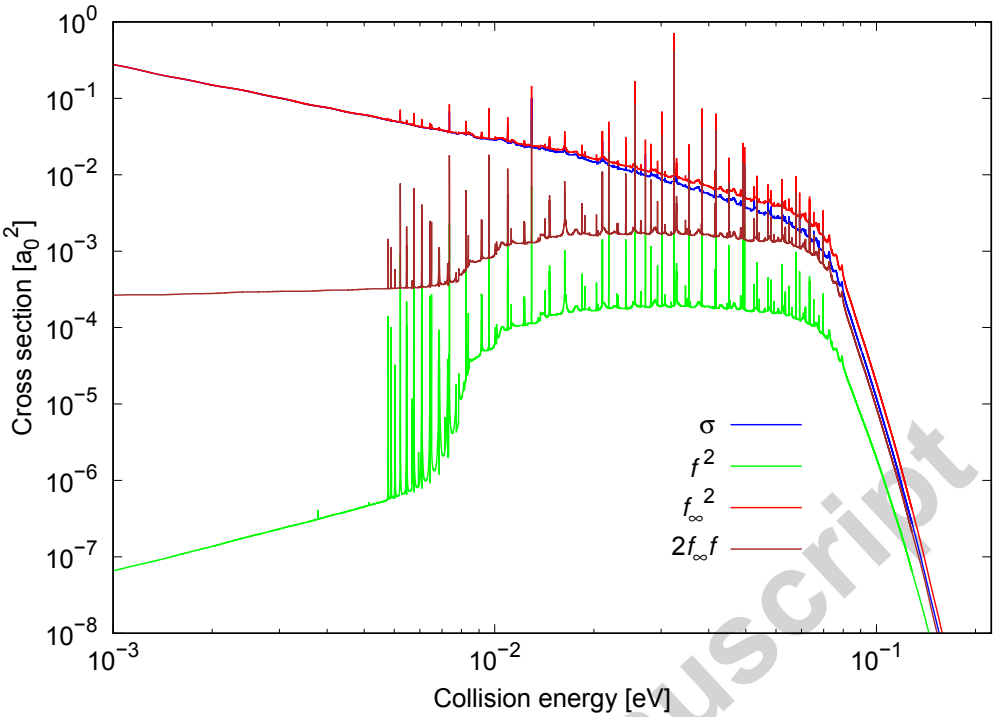


Figure 6: Contributions from three terms $|\chi_{E,q'} f \psi_{v'',q''}|^2$, f_∞^2 $|\chi_{E,q'} \psi_{v'',q''}|^2$, and $|2\ f_\infty\ \chi_{E,q'} f \psi_{v'',q''}|$ to cross-sections $\sigma^{1\to 0}(E;0)$ for spontaneous radiative association of $HeLi^+$ within the $1 \to b$ process when the transition dipole-moment function is represented as $d_x(R) = f(R) + f_\infty$, where $f_\infty = -2.537812\ ea_0$ is a constant. Only those resonances with a width larger than 0.01 cm^{-1} are shown.

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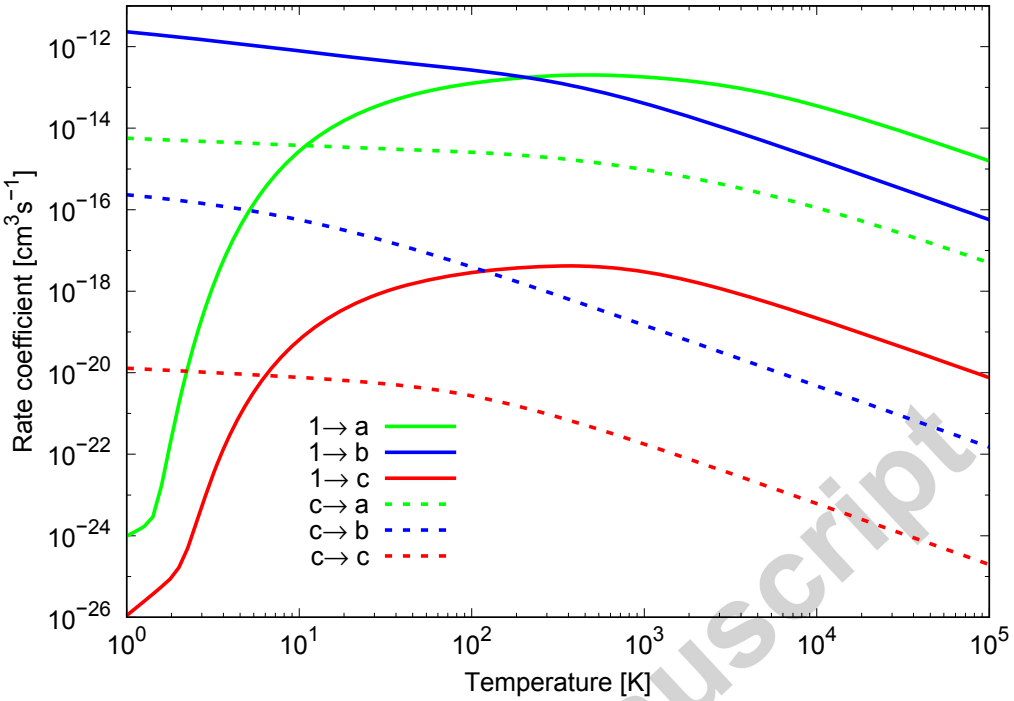


Figure 7: Rate coefficients $\alpha(T;0)$ of the $\Pi \rightarrow \Sigma$ processes (solid lines) calculated in this work and of the $\Sigma \rightarrow \Sigma$ processes (dashed lines) calculated in Ref. [25] for spontaneous radiative association of $HeLi^+$.

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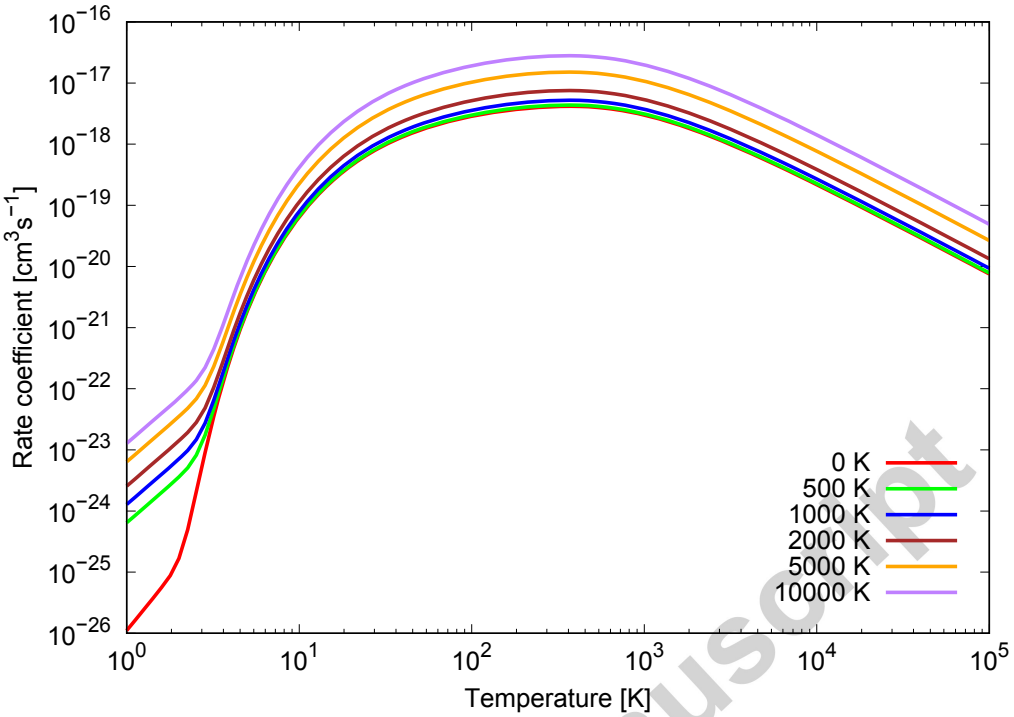


Figure 8: Rate coefficients $\alpha(T;T_\mathrm{b})$ at different background temperatures for the stimulated radiative association $1 \to c$ of $HeLi^+$.

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Table 5: Values of the Kooij-function parameters representing the rate coefficients for the spontaneous radiative association of He(2^3P) with lithium cations.[a]

process	T (K)	α (cm^{3}s^{-1})	β	γ (K)
$1 \rightarrow a$	10-280	2.3111(-13)	0.2212	37.7188
	280-3000	5.3264(-13)	-0.6370	313.9012
	3000-500000	9.4571(-12)	-1.4927	3377.3163
$1 \rightarrow b$	10-280	1.4451(-13)	-0.5635	3.0826
	280-3000	2.8869(-13)	-1.4353	233.6361
	3000-500000	3.4698(-13)	-1.4999	359.5807
$1 \rightarrow c$	10-280	4.9238(-18)	0.1523	39.2179
	280-3000	2.3113(-17)	-1.2555	534.7395
	3000-500000	4.6345(-17)	-1.4997	1011.1481

[a] $x(-y) \equiv x \times 10^{-y}$.

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